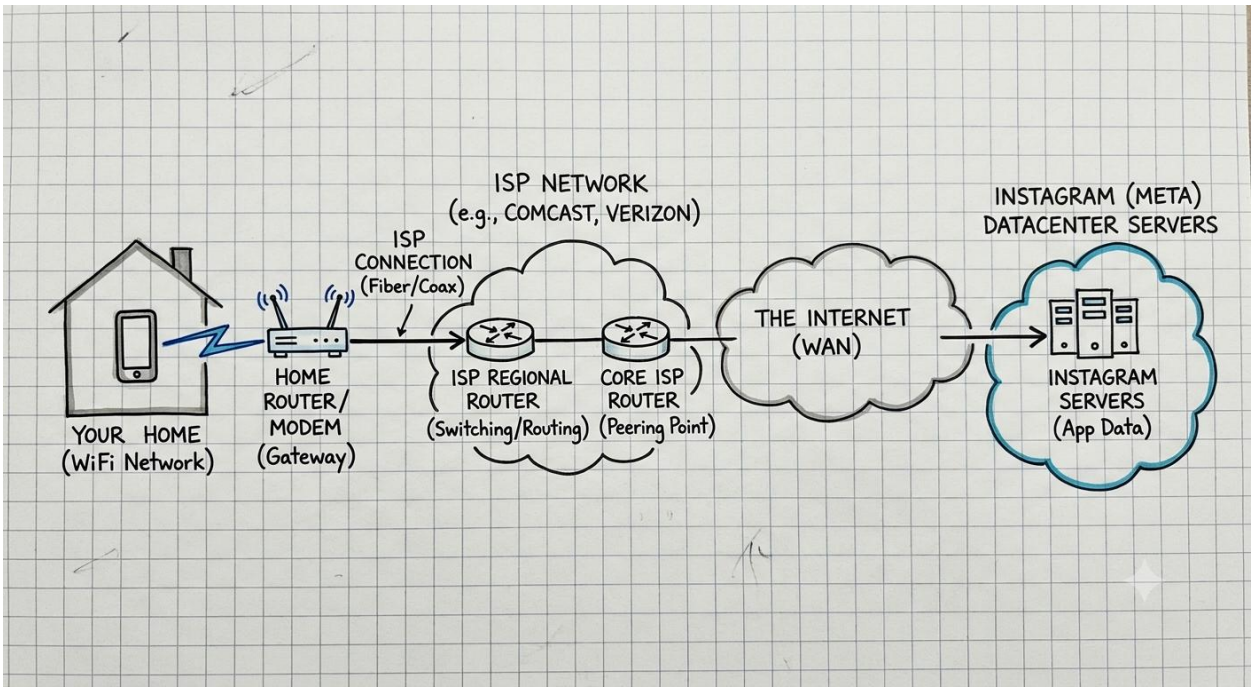
1)Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

Ans.



2)Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

Ans

WAN Technology	**Speed**	**Cost**	**Real-Life Example & Use Case**
Broadband Internet (Fiber, Cable, DSL)	**Fast** *(Good for everyday apps)*	**Low** *(Affordable monthly bill)*	**Your Home** Used for streaming videos, scrolling social media, and casual web browsing.
MPLS (Multi-Protocol Label Switching)	**Reliable & Stable** *(Consistent speeds)*	**High** *(Costs more because it's private)*	**Zomato Office** Used to safely connect different office branches together to share company files without lag.
Leased Line (Dedicated Fiber Line)	**Ultra-Fast & Shared with No One**	**Very High** *(Most expensive option)*	**IRCTC Ticketing Server** Used where millions of people are booking tickets at once and the system cannot afford to crash or slow down.

3)Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop. Focus on how the VPN creates a private tunnel over the public WAN.

Ans

College Lab Security Checklist

1. **Secure physical hardware with cable locks and room access**
    - Bolt computers to desks and keep the lab locked when staff isn't present to prevent theft or tampering.

2. **Use reboot-to-restore software (like Deep Freeze)**
    - Ensures any malware, downloaded files, or system changes made by students are wiped clean every time the PC restarts.

3. **Isolate the lab network using VLANs**
    - Place lab computers on a separate network segment so students cannot access sensitive campus administration or financial servers.

4. **Require unique individual student logins**
    - Avoid shared generic guest accounts so you can track exactly who was logged into a machine if a security incident occurs.

5. **Restrict USB ports and external storage devices**
    - Use Group Policy to block unauthorized flash drives or external hard drives that could carry malware or steal data.

6. **Disable or secure unused Ethernet wall ports**

    - Prevent unauthorized devices or rogue routers from being plugged directly into the active campus network.

7. **Automate updates and monitor network traffic**

    - Set up automatic security patches for operating systems and software, and keep an eye out for unusual bandwidth spikes.

4)Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

Ans. For a large e-commerce company like Flipkart connecting major warehouses in **Ahmedabad, Mumbai, and Delhi**, the best WAN technology to use is **SD-WAN (Software-Defined Wide Area Network)**.

1. **Lower Costs & More Flexibility**
    - **Why it helps:** SD-WAN can use affordable everyday internet connections (like high-speed fiber broadband) instead of relying only on expensive, rigid private lines. This saves a lot of money when connecting large warehouses across different cities.

2. **Smart Traffic & Fast Data Speed**
    - **Why it helps:** Warehouses need real-time inventory updates so items don't get oversold. SD-WAN automatically picks the fastest, most reliable path for data traffic, ensuring that the systems in Mumbai, Delhi, and Ahmedabad stay instantly synced without lag.